KERALA STATE ELECTRICITY REGULATORY COMMISSION
THIRUVANANTHAPURAM

Petition No: OP 84/2023

Present	:	**Sri. T.K. Jose, Chairman** **Adv. A. J. Wilson, Member** **Sri. B. Pradeep, Member**
In the matter of	:	Petition filed by M/s Technopark for the Truing up of Accounts for the financial year 2022-23
Petitioner	:	M/s Technopark, Trivandrum
Represented by	:	Sri. Madhavan Praveen, GM Projects Sri. Viswanathan N, Deputy Manager (Finance) Sri. Anfal, Deputy Manager (Electrical)
Respondent	:	M/s Kerala State Electricity Board Limited
Represented by	:	Sri. Ajith Kumar K.N, Executive Engineer, KSEB Ltd Sri.Shan B.S, Asst. Executive Engineer, KSEB Ltd Smt. Archana M, Asst. Engineer, KSEB Ltd
Date of Hearing	:	27.02.2024

ORDER DATED 26.03.2025

1. M/s Technopark, Thiruvananthapuram, is a deemed distribution licensee for distribution of electricity in the Technopark campus. M/s. Technopark has four licence areas namely (1) Technopark Phase I, (2) Technopark Phase II & III, (3) Technocity Pallippuram and (4) Technopark Kollam.

2. The Commission had notified the KSERC (Terms and Conditions for Determination of Tariff) Regulations, 2021 for the five-year Control Period starting from the first day of April 2022 and ending on the thirty first day of March 2027. M/s Technopark has filed the petition for truing up of accounts for the year 2022-23 as per the provisions of the KSERC (Terms and Conditions for determination of Tariff) Regulations 2021. The present petition is for the first year of the Control Period. The Commission vide Order in OP No.12/2023 dated 28.02.2024 has approved the ARR&ERC for the Control Period. In the present petition for truing up the accounts, the licensee has claimed a revenue surplus of Rs.250.37 lakh compared to revenue surplus of Rs.641.72 lakh approved by the Commission in the Order on ARR&ERC. A comparison of the claim made in the present petition with the ARR&ERC approved by the Commission is tabulated below.

Table 1
Comparison of ARR & ERC and Truing Up for 2022-23 (Rs. Lakh)

Particulars	ARR & ERC Projected	ARR & ERC Approved	Petition for truing up
Cost of power purchase/ generation	4652.88	4652.67	4652.88
Employee Expenses	53.30	27.14	54.54
Administration and General	93.71	36.09	92.05
Repairs and Maintenance	449.91	317.37	447.65
Depreciation	171.89	142.60	162.45
Interest on long term loans	9.03	7.87	8.96
Interest on consumer deposits	72.70	63.91	72.70
Return on Net Fixed Assets	52.43	21.07	86.60
Total Expenditure/ARR	**5555.86**	**5268.72**	**5577.83**
Revenue from sale of power	5751.09	5751.09	5750.11
Non-tariff income	90.54	159.35	78.09
Total Revenue / ERC	**5841.63**	**5910.44**	**5828.20**
Net Revenue Surplus/(Deficit)	**285.77**	**641.72**	**250.37**

Table 2
Comparison of Income and Expenditure for the previous years **(Rs. lakh)**

Particulars	2020-21		2021-22		2022-23
	Truing Up Petition	Trued Up	Truing Up Petition	Trued Up	Truing Up Petition
Purchase of Power	3511.93	3466.17	3494.83	3443.50	4652.88
Prior Period Power Purchase Cost	-	9.71	-	-	-
Employee Cost	45.85	41.04	50.11	43.03	54.54
R&M Expenses	376.29	293.47	404.85	329.87	447.65
A &G Expenses	89.27	57.35	92.03	62.95	92.05
Depreciation	154.06	142.60	162.68	142.60	162.45
Interest and finance charges on long term loans	30.08	17.32	24.96	7.87	8.96
Interest on consumer security deposits etc	98.85	98.85	98.80	98.80	72.70
Return on NFA	48.96	25.35	53.08	21.07	86.60
Total Expenditure	**4355.30**	**4151.86**	**4381.34**	**4149.69**	**5577.83**
Revenue from Sale of Power	4260.79	4260.79	4317.28	4317.28	5750.11
Non-Tariff Income	94.04	171.49	83.67	162.14	78.09
Total Income	**4354.83**	**4432.28**	**4400.95**	**4479.42**	**5828.20**
Net Surplus/(Deficit)	**(0.47)**	**280.42**	**19.61**	**329.73**	**250.37**

3. The Commission admitted the petition as **OP No. 84/2023.**

Hearing on the Petition

4. The hearing on the petition was conducted at the Court Hall, Office of the Commission on 27-02-2024. The petitioner, M/s Technopark was represented by Shri. Madhavan Praveen – G.M. Projects, Shri. Anfal - Deputy Manager (Electrical) and Shri. Viswanathan – Deputy Manager (Finance). Shri. Anfal, presented the details of the petition on the truing up of accounts for the year 2022-23. The important submissions made by Technopark during the hearing are: -

a) The petitioner has served a total of 933 consumers in financial year 2022-23 and the sale during the year is 643.34 lakh units.
b) The distribution loss for the year is 4.42%.
c) The power purchase cost for 669.67 lakh units purchased from KSEB Ltd is Rs.4652.88 lakh. The total energy requirement for the licensee is 669.67 lakh units. Out of the total energy requirement, 3.40 lakh units is generated from the four rooftop solar plants with total capacity of 285 kW.
d) The Employee Cost of Rs.54.54 lakh includes the salary and benefits paid to the permanent employees of Technopark directly involved in the distribution business activity.
e) The A&G Expenses of Rs.92.05 lakh claimed by the licensee includes Payment of Rs.36.98 lakh towards Electricity Duty under Section 3(1) of the Kerala Electricity Duty Act 1963.
f) The R&M Costs for the year is Rs.447.65 lakh which is the cost for operation, maintenance and repairs of the distribution system carried out by licensed electrical contractors and the work was awarded after a due transparent process of competitive bidding.
g) The licensee stated that, there was no addition to the fixed assets during the year. There is a capital expenditure of Rs 158.27 lakh towards various on-going capital works
h) The depreciation claimed by the licensee in the petition for truing up for the year 2022-23 is Rs.162.45 lakh.
i) The interest and finance charges claimed by the licensee for the year 2022-23 amounts to Rs.81.66 lakh which includes interest on long term loans and interest on consumer security deposit.
j) The licensee claimed Return on Net Fixed Assets amounting to Rs.86.60 lakh worked out at 5.50% on the NFA of Rs.1574.46 lakh at the beginning of 2022-23.
k) The revenue earned from sale of power was Rs.5750.11 lakh and the Non-tariff income is Rs.78.09 lakh

l) Accordingly, the total Income towards Revenue from sale of power and Non-Tariff receipts for the year 2022-23 is Rs.5828.20 lakh and the total expenditure is Rs.5577.83 lakh. Thus, the licensee has claimed a revenue surplus of Rs.250.37 lakh.

5. KSEB Limited, the power seller Licensee to Technopark, vide letter dated 26.02.2024 furnished the following counter statement/remarks on the truing up petition for the year 2022-23 filed by Technopark.

a) To disallow the cost incurred for additional power procurement due to higher distribution loss.
b) To limit the O&M expenses within the norms specified as the O&M expenseis higher than the approved ARR norms.
c) To take a uniform approach regarding Section 3(1) duty as the same is claimed by Technopark.
d) Deprecation for the year may be allowed only after prudence check.
e) RoNFA may be allowed only after verifying the details with regard to asset addition.
f) The interest on accumulated surplus need to be treated as part of non-tariff income.
g) The purchase units and the power purchase cost tally with the records of KSEB Ltd.

Analysis and decision of the Commission

6. The Commission has carefully considered the petition for Truing Up of Accounts for the year 2022-23, the counter statements furnished by the KSEB Ltd and the views presented by the licensee during the hearing. The analysis and decisions of the Commission on the petition for truing up of accounts for the Financial Year 2022-23 are detailed below.

Energy sales and consumer mix

7. The licensee has in the petition submitted area wise details of the number of consumers and sales. During 2022-23, there were a total of 933 consumers compared to 862 consumers in the previous year 2021-22. The total energy sale as per the petition for Truing Up of Accounts for 2022-23 is 643.34 lakh units. The actual sales unit for the previous year 2021-22 was 500.77 lakh units. A comparison of the actual sale of power and category wise number of consumers for the year 2021-22 and 2022-23 is given below.

Table 3
Comparison of Energy Sales and No of Consumers for 2021-22 and 2022-23

Particulars	2021-22		2022-23	
	Number of consumers	Sales (Lakh Units)	Number of consumers	Sales (Lakh Units)
LT – IV B IT/ITES below 10kW	224	9.07	220	10.93
LT – IV B IT/ITES above 10kW up to 20kW	70	4.59	74	7.01
LT – IV B IT/ITES above 20kW	242	210.6	238	249.31
LT - VI (A) General	3	0.12	2	0.39
LT VI (B) General	9	1.04	9	0.59
LT VI (c) General (1ph or 3 ph)	34	2.25	33	2.38
LT VI (F) General (1 ph)	44	0.66	51	0.68
LT VII (A) Commercial (1-Ph)	51	0.26	50	0.44
LT VI (F) General (3 ph)	57	14.58	125	22.86
LT VII (A) Commercial (3-Ph)	59	1.80	57	3.77
LT VII (B) Commercial	11	0.02	11	0.02
LT IV A Industrial below 10 KW	6	0.16	6	0.22
LT IV A Industrial above 10 KW below 20 KW	2	0.13	4	0.38
LT IV A Industrial above 20 KW	6	2.38	7	2.82
LT VI (G) General	1	0.10	1	0.07
LT IX Display Lighting and Hording	2	0.09	2	0.08
Self-Consumption	2	1.96	2	2.74
Street Lighting	16	1.51	18	1.46
Sub Total (LT)	**839**	**251.31**	**910**	**306.15**
HT -I (B) IT & IT Services	17	232.97	17	313.82
HT -II (A) General	2	10.89	2	13.98
HT II B	3	2.79	3	5.40
HT- IV Commercial	1	2.82	1	3.99
Sub Total (HT)	**23**	**249.46**	**23**	**337.19**
Grand Total	**862**	**500.77**	**933**	**643.34**

8. The major sale of the licensee is to LT IV B and HT I B category of consumers. As compared to last year there is an increase in the no.of consumers as well as the sales units. The increase is mainly due to new consumers in LT VI (F) category. The total energy sold to these consumers was 643.34 lakh units in FY 2022-23 compared to 500.77 lakh units in FY 2021-22. ***After analyzing the details and duly considering the consumption pattern, the Commission hereby approves the actual energy sales of 643.34 lakh units as claimed in the petition for Truing Up of Accounts for the year 2022-23.***

Energy Requirement and Distribution loss

9. As per the petition, the total energy requirement for the year is 673.07 lakh units, which is inclusive of energy purchased from KSEB Ltd 669.67 lakh units and 3.40 lakh units from the solar plant. The actual distribution loss for the year is 4.42% as against the distribution loss of 3.45% approved by the Commission in the Order on ARR&ERC. As compared to the previous year 2021-22 the actual distribution loss have been increased from 3.45% to 4.42%. The details of the distribution loss as per the petition and previous year are as shown below.

Table 4
Distribution loss for the period 2021-22 and 2022-23

Particulars	2021-22		2022-23
	As per Truing Up Petition	Trued Up	As per Truing Up Petition
Energy sales (lakh units)	500.77	500.77	643.34
Energy Requirement (lakh units)	518.65	511.09	673.07
Distribution loss (lakh units)	17.88	10.32	29.73
Distribution loss (%)	**3.45%**	**2.02%**	**4.42%**

10. Commission while truing up the accounts for the previous years of the control period had approved 2.02 % as the distribution loss for each year. The licensee has submitted the phase wise distribution loss of the licensee. The phase wise distribution loss for the year 2021-22 and 2022-23 is as given below.

Table 5
Phase wise Distribution loss for 2022-23

Particulars	Solar Generation	Energy purchase	Energy Sales	Distribution Loss	Loss (%)
	(lakh unit)	(lakh unit)	(lakh unit)	(lakh unit)	
Phase -1	0.157	432.09	410.41	21.84	5.05%
Phase - 2&3	3.24	212.75	209.75	6.25	2.89%
Phase-4 Technocity	0	18.58	17.19	1.39	7.49%
Phase-5 Kollam	0	6.24	5.99	0.25	4.04%
Overall Losses	**3.40**	**669.67**	**643.34**	**29.73**	**4.42%**

Table 6
Phase wise Distribution loss for 2021-22

Particulars	Solar Generation	Energy purchase	Energy Sales	Distribution Loss	Loss (%)
	(lakh unit)	(lakh unit)	(lakh unit)	(lakh unit)	
Phase -1	0.16	352.53	340.04	12.65	3.59%
Phase - 2&3	3.94	147.18	147.86	3.26	2.16%
Phase-4 Technocity	0	10.34	8.68	1.66	16.07%
Phase-5 Kollam	0	4.5	4.2	0.3	6.73%
Overall Losses	**4.10**	**514.55**	**500.77**	**17.88**	**3.45%**

11. It is seen that though there is reduction in the loss in phase 4 and phase 5, the distribution loss in Phase I and 2 & 3 are increasing consistently. With regard to high distribution loss, the licensee has stated the following;

    i. *The loss during the year has increased despite the petitioner's best efforts to keep the consumer meters accurate.*
    ii. *There is an increase in consumption as major companies are resuming their operations after COVID 19 pandemic.*
    iii. *Many of the transformers operated under load, causing increased fixed losses, and an investigation is underway.*
    iv. *Some of the CT installed in the campus are old tape wounded type and low accuracy class, which had effect on the loss in technopark.*

12. Distribution loss is a controllable parameter as per Tariff Regulations, 2021. Regulation 73(3) clearly specifies the methodology to be considered to account for any variation between the actual level of distribution loss and the approved level of the distribution loss. The relevant portions of the Regulation are as shown below:

73. Distribution loss...
.....

(3) Any variation between the actual level of distribution loss and the approved level of the distribution losses shall be dealt with, as part of the truing up of the respective financial year in the following manner:-

i. if the actual distribution loss is higher than the approved level of distribution loss for any particular financial year of the control period, then the quantum of power purchase corresponding to the excess distribution loss for that financial year shall be disallowed at the average cost of power purchase for the respective financial year;

ii. if the actual distribution loss is lower than the approved level of distribution loss for any particular financial year of the control period, then the savings in the power purchase cost corresponding to the difference in distribution loss for that financial year at the average cost of power purchase for the respective financial year, shall be shared between the distribution business/licensee and the consumers in the ratio of 2:1

13. Accordingly, as per the provisions of the Tariff Regulations, distribution loss is a controllable performance parameter. As per regulation 73(3) of the Tariff Regulations, 2021, if actual distribution loss is higher than the approved level, the quantum of power purchased corresponding to the excess distribution loss shall be disallowed at the average power purchase cost. ***The Commission hereby decides to limit the distribution loss to the approved level of 3.45% for the purpose of truing up. The licensee is directed to take effective***

measures to limit the distribution loss at the approved level. Accordingly, based on the approved distribution loss, the approved energy requirement for the year 2022-23 is tabulated below.

Table 7
Distribution loss approved for the year 2022-23

	Particulars	As per Truing Up Petition	Trued Up
A	Total Energy Output (lakh units)	643.34	643.34
B	Distribution loss (%)	4.42%	3.45%
C	Distribution loss (in lakh units) [A/100-B)*B]	29.73	22.99
D	Total Energy Requirement (lakh units)	673.07	666.33
E	Energy Generated from Solar	3.40	3.40
F	Energy Required from KSEB Ltd (lakh units)	669.67	662.93
Excess Distribution Loss in lakh units (669.67-662.93)			**6.74**

Solar Generation

14. The solar generation from the four solar plant units having capacity 200 kWp, 50 kWp, 25 kWp and 10 kWp owned by the petitioner is given below.

Table 8
Details of Solar Generation over the years

Particulars	2020-21		2021-22		2022-23	
	CUF Achieved	Units Generated	CUF Achieved	Units Generated	CUF Achieved	Units Generated
200 kWp	16.15 %	282944	16.58 %	290475	12.78%	223883
50 kWp	16.30 %	71411	15.28 %	66940	15.05%	65923
25 kWp	16.80 %	36787	16.62 %	36408	15.63%	34224
10 kWp	18.07 %	15828	18.12 %	15869	17.91%	15693
Total Generation		406970		409692		339723
	4.07 lakh units		**4.10 lakh units**		**3.40 lakh units**	

** CUF = Yearly Generation x 100 / (Plant Capacity x 24 x 365)*

15. The licensee has stated that the Capacity Utilisation Factors (CUF) varies from 12.78 % to 17.91%. The Inverter of the 200-kW solar plant had developed technical issues several times and was got repaired each time. The solar power plants at Technopark were installed during 2018-19 and the installed panels are of the polycrystalline type with a panel efficiency of 16-17% only which is stated to be the reason for lower generation. Commission takes note of the fact that the generation of the plants is significantly low compared to the previous years. The licensee is to note that the CUF of the solar plants are lower than levels specified in Regulation 54(3) of Kerala State Electricity Regulatory Commission (Renewable Energy and Net Metering) (First Amendment) Regulations, 2022. As per regulation, the Capacity Utilization Factor for Solar PV project shall be 19%. The licensee is directed to properly verify the reasons for the reduction in generation and to make necessary steps to maximize the generation.

Power Purchase Cost

16. As per the petition, the total energy required for serving the consumers was 673.07 lakh units for the year 2022-23 as against approved 666.19 lakh units. The petitioner generated about 3.40 lakh units of electricity from the four rooftop solar plants of total capacity 285 kW. The balance requirement of 669.67 lakh units was met from the energy purchased from the State Electricity Utility, KSEB Ltd at a total cost of Rs. 4652.88 lakh as against approved Rs. 4652.67 lakh. The average per unit cost of power purchase was Rs. 6.95/unit for 2022-23. As compared to the previous year, the actual power purchase cost has increased. The split up of power purchase cost for the year 2022-23 and previous year is tabulated below.

Table 9
Phase wise power purchase cost for the period 2022-23

Campus	Maximum demand kVA	Energy Received lakh units	Fixed charges Rs lakh	Variable charges Rs lakh	Incentive Rs. lakh	Other charge Rs. lakh	Total charge Rs. lakh
Phase-1	1,35,000	432.09	500.25	2,541.91	(38.48)	0	3,003.68
Phase -2 & 3	55,158	212.75	205.00	1,252.55	(19.78)	0	1,437.77
Phase -4 Technocity	9,076	18.58	33.64	109.81	-	11.83	155.28
Phase -5 Kollam (2)	3,600	6.24	14.17	41.35	(0.36)	1.00	56.16
Total	2,02,834	669.67	753.05	3,945.62	(58.62)	12.83	4,652.88
Average Power Purchase Cost (Rs /per unit)							**6.95**

Table 10
Phase wise power purchase cost for the period 2021-22

Campus	Maximum demand kVA	Energy Received lakh units	Fixed charges Rs lakh	Variable charges Rs lakh	Incentive Rs. lakh	Other charge Rs. lakh	Total charge Rs. lakh
Phase-1	1,35,000	352.53	459.00	1,938.94	-	0	2,397.94
Phase -2 & 3	45,216	147.18	153.73	809.48	-	0	963.21
Phase -4 Technocity	9,000	10.34	30.6	56.86	-	5.94	93.40
Phase -5 Kollam (2)	3,600	4.50	12.24	27.88	(0.41)	0.56	40.27
Total	**1,92,816**	**514.55**	**655.57**	**2,833.16**	**(0.41)**	**6.51**	**3,494.83**
Average Power Purchase Cost (Rs /per unit)							**6.79**

17. As mentioned in previous paragraphs, as per the Tariff Regulations 2021, if the actual distribution loss is higher than the approved level of distribution loss, then the quantum of power purchased corresponding to the excess distribution loss shall be disallowed at the average cost of power purchase. ***Accordingly, the power purchase cost based on the approved distribution loss of 3.45% for the year 2022-23 is as shown below.***

Table 11

Approved power purchase cost for 2022-23

1	Actual distribution loss (In lakh units)	29.73
2	Approved distribution loss (In lakh units)	22.99
3	Excess distribution loss (lakh units) (1-2)	6.74
4	Average Power Purchase Cost (Rs. per unit)	6.95
5	Excess power purchase cost to be disallowed (Rs. lakh) (4) * (3)	46.84
6	Actual Power purchase cost (Rs. lakh)	4,652.88
7	**Approved Power purchase cost (Rs. lakh) (6-5)**	**4,606.04**

Operation & Maintenance Expenses

18. Operation and Maintenance expenses are controllable expenses which include Employee cost, Repair & Maintenance expenses and Administrative & General expenses. The Commission while approving the ARR & ERC for the control period had approved the O&M expenses as per the normative level and had stated that any additional expenses relating to uncontrollable factors or one-time expenses can be considered at the time of Truing Up of accounts only after prudence check by the Commission. The Commission while approving the norms in the Tariff Regulations, 2021 had approved the Operation and Maintenance expense of Rs. 380.60 lakh for 2022-23. The licensee in the petition has claimed total operation and maintenance expense of Rs.594.24 lakh. The comparison of O&M expenses with the previous year is tabulated below.

Table 12

Comparison of O&M expenses (Rs. lakh)

O&M Expenses	2021-22			2022-23	
	Norms	As per Truing Up Petition	Trued up	Norms	As per Truing Up Petition
Employee expenses	34.54	53.85	43.03	27.14	54.54
R&M expenses	210.73	404.85	329.87	317.37	447.65
A&G expenses	60.19	92.03	62.95	36.09	92.05
Total	**305.46**	**550.73**	**435.85**	**380.60**	**594.24**

Employee cost

19. The actual employee cost claimed by the licensee in the petition is Rs. 54.54 lakh as against the approved norms of Rs. 27.14 lakh. The actual employee is higher compared to the norms. The major items booked under Employee Cost are Consolidated pay, salaries, dearness allowances, HRA, other allowance etc. The employee cost claimed for the previous year 2021-22 was

Rs. 53.85 lakh. The Commission while truing up had approved an amount of Rs. 43.03 lakh. The split up details and a comparison of the employee cost for the current year 2022-23 and previous year 2021-22 is shown below.

Table 13
Comparison of Employee cost (Rs. lakh)

	2021-22	2022-23
Particulars	**Truing Up claim**	**Truing Up claim**
Basic Salary	15.89	17.24
Dearness Allowance (DA)	5.72	6.21
House Rent Allowance	0.53	0.59
Other Allowances (CCA)	0.09	0.11
Personal Pay	0.14	-
Arrears	0.79	-
Consolidated Pay	30.67	30.39
Grand Total	**53.85**	**54.54**

20. The licensee for the distribution business has claimed the employee cost on apportionment basis. The ratio for apportionment and the employee cost is tabulated below.

Table 14
Apportionment of Employee Cost for 2022-23

Designation	Posts	Gross Salary Rs.lakh	Ratio	Apportioned Salary Rs.lakh
Chief Executive Officer	1.00	33.66	10%	3.37
Chief Finance Officer	1.00	22.32	20%	4.46
GM (Projects)	1.00	20.00	25%	5.00
Deputy Manager (Electrical)	1.00	10.97	100%	10.97
Deputy Manager Finance	1.00	17.24	40%	6.90
Sr. Executive Finance	1.00	4.64	75%	3.48
Apprentice Trainee	1.00	2.55	50%	1.28
Jr. Electrical Engineer	1.00	1.28	100%	1.28
Supervisors/Meter readers	4.00	19.01	75%	14.26
Civil Engineer	1.00	3.74	25%	0.94
Total		**135.41**		**54.54**

21. It is seen that the employee cost claimed in the petition is Rs. 54.54 lakh. The Commission has examined the details furnished by the licensee and the claim in the Truing up of accounts as per the provisions of the Tariff Regulations, 2021. As per the Tariff Regulations, 2021, the licensee shall be allowed to recover the operation and maintenance expenses as per the norms specified in Annexure-7 of Tariff Regulations for each financial year of the Control Period. The Annexure-7 of Tariff Regulations 2021 specifies, the base year (2021-22) norms as per Table-9 and approved norms (Fifth Table under para

3(iii) of Annexure-7) of the O&M expenses of Technopark. The details are given below.

Table 15

O&M Expenses as per the Tariff Regulations, 2021 (Rs. lakh)

Particulars	Base year	Control period				
	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27
Employee Cost	25.99	27.14	28.35	29.61	30.93	32.31
R&M Expenses	303.84	317.37	331.51	346.28	361.70	377.81
A&G Expenses	34.55	36.09	37.69	39.37	41.13	42.96
Total	**364.37**	**380.60**	**397.55**	**415.26**	**433.76**	**453.08**

22. The paragraph-1 of the Annexure-7 of the Tariff Regulations, 2021 specifies inflation rate adopted for tariff determination for the Control Period from 2022-23 to 2026-27. Further, the 'Note' to the paragraph-1 to Annexure-7 permits that, the actual inflation will be allowed at the stage of truing up for the respective years. The relevant provisions of the Regulations, 2021 is stated below.

"Methodology adopted for specifying the Normative Operation & Maintenance (O&M) Costs for the 5-year Control Period (2022-27)

1. The O&M costs consist of:
   - *Employee cost;*
   - *Administrative & General (A&G) expenses; and*
   - *Repair & Maintenance (R&M) costs.*

The annual inflation/ escalation rate under the Control Period (2018-22) was fixed at 4.84% based on the CPI (Industrial workers) & WPI indices in the weightage ratio 70:30, for the 4-year period (2013-14 to 2016-17). Considering the 5-year MYT period in this Regulation and the subdued performance of the economy in the year 2020-21 due to the 'Covid 19' pandemic; it is proposed to fix the annual escalation factor for the 5-year MYT Control Period (2022-23 to 2026-27) as follows: -

a) *The average of the increase in the CPI & WPI indices, for the 4 year period (2017-18 to 2020 -21), as given in Table- 1 below:*

Computation of Escalation Factors for this MYT Regulation

Sl.No	*Escalation factors/ year*	*2017-18*	*2018-19*	*2019-20*	*2020-21*
1	*CPI (Industrial Workers) 2001=100*	*284.42*	*299.90*	*322.50*	*338.69*
2	*Annual CPI increase %*	*3.08*	*5.44*	*7.54*	*5.02*
3	*WPI (2011-12) series*	*114.90*	*119.80*	*121.80*	*123.40*
4	*Annual WPI increase %*	*2.96*	*4.26*	*1.67*	*1.31*
5	*CPI:WPI (70:30) weightage (%)*	*3.04*	*5.09*	*5.78*	*3.91*
6	*Average 4 Year annual increase @ **70:30 weightage (%)***	***4.454%***			

Note:

The Commission notes that there has been severe fluctuations as far as the inflation indices are concerned. Further, due to the outbreak of COVID 19, and the overall depressing economic situation faced by all major industries in the country, the offtake of electricity during this period has considerably reduced and there is a glut as far as the electricity market is concerned.

However as of now, the economy seems to be on a path of revival and the increase in oil prices have also added to the overall inflation in the economy. Hence, the Commission feels that any data of past indices may not completely reflect the true economic situation in the country and its inflationary trends and therefore is of the opinion that, for the purpose of ARR&ERC for the Control Period, it would be advisable to rely on historical inflation data for the previous four years. But, the actual escalation rates based on the actual inflation existing for each year of the Control Period shall be considered at the time of the truing up of the concerned financial year and would be a more reliable basis. Hence, the actual inflation indices will be allowed at the stage of truing up for the respective years."

23. Accordingly, as discussed in the preceding paragraphs and as specified under Annexure-7 of the Tariff Regulations, 2021, the base year O&M expenses for the Year 2021-22 is Rs. 364.37 lakh. The average escalation rate approved by the Commission for the Control Period from 2022-23 to 2026-27 based on the average of the 'consumer price index and wholesale price index' is 4.454%. By applying the escalation rate of 4.454% on the O&M expenses of the base year 2021-22, the O&M Expenses for 2022-23 is Rs. 380.60 lakh. However, the Note to the paragraph-1 to the Annexure-7 of the Tariff Regulations, 2021 provides that, the escalation rates based on the actual inflation indices shall be adopted for approving the O&M expenses while truing up of the respective years. The actual inflation rate of the year 2022-23 is arrived as detailed below.

Table 16
Actual inflation for the year 2022-23

Sl No	**Particulars**	**2021-22**	**2022-23**
1	Consumer Price Index (2016 price level)	123.6333	131.1167
	Increase in CPI in 2022-23 over 2021-22		6.05%
2	Wholesale Price Index (base year 2011-12)	139.4	152.5
	Increase in WPI in 2022-23 over 2021-22		9.40%
3	Weighted average increase (70% CPI & 30%WPI)		**7.06%**

24. Based on the actual escalation of 7.06% in the year 2022-23, the revised O&M expenses of Technopark is approved by the Commission considering actual inflation as given below;

Table 17

Revised O&M expenses of Technopark approved for the year 2022-23 considering Actual Inflation (Rs. lakh)

Particulars	Base year	Control Period- 2022-23	
	2021-22	Actual inflation	O&M Approved
Employee Cost	25.99	7.06%	**27.82**
R&M Expenses	303.84	7.06%	**325.29**
A&G Expenses	34.55	7.06%	**36.99**
Total	**364.37**		**390.10**

25. Considering the above, ***the Commission hereby approves the employee cost of Rs. 27.82 lakh for the year 2022-23 as per the norm after adjusting actual inflation.***

Table – 18

Employee cost approved for the year 2022-23 (Rs lakh)

Particulars	As per Regulations	As per Truing Up Petition	Trued Up
Employee Cost	27.14	54.54	27.82

R & M Expenses

26. The actual R&M expense claimed by the licensee in 2022-23 is Rs. 447.65 lakh as against the normative amount of Rs. 317.37 lakh specified in Tariff Regulations, 2021. The licensee has stated that the R&M of the distribution system in Technopark license area is carried out by licensed electrical contractors. The contract is awarded after a due transparent process of competitive tendering as per the existing rules and government directions. The tenders for R&M works are invariably invited through the government e-tendering portal. Thus, it is ensured that the contract amount discovered is the lowest possible price.

27. During the previous year, the claim was 404.85 lakh. The Commission after prudence check approved Rs. 329.87 lakh. The comparison of the R&M expense of current petition and previous year is tabulated below:

Table 19

Details of R&M expenses for the years 2021-22 and 2022-23 (Rs. lakh)

Particulars	2021-22		2022-23
	As per Truing Up Petition	Trued up	As per Truing Up Petition
Phase-1 (Plant & Mach Only)	254.15	178.71	269.29
Phase 2 & 3 (do)	91.42	92.51	68.25
Phase -5 Kollam (do)	33.26	32.63	32.97
Phase 4 Technocity do)	-	-	30.38
Material and Spares Purchased	10.80	10.80	11.16
One-Time / Break-down works carried out	15.22	15.22	35.60
Gross R&M Expenses	**404.85**	**329.87**	**447.65**

28. There is an overall increase of 10.57 % in this cost compared to previous year, which is inclusive of new distribution area under Phase-IV in Pallippuram which amounted to Rs 30.38 lakh.

29. The licensee in the petition states that the R&M contracts were awarded after a fair and transparent process of on-line tendering where there were a fair number of competitive bids for the contract and that there is little scope for reducing these costs. There is no way to compensate for the substantial gap between the actual R&M cost genuinely incurred and that allowed in the ARR. The licensee also stated that this difference in cost does not in any way reflect any defect or inefficiency on the part of the petitioner. The cost incurred is the lowest discovered through transparent bidding. The licensee has submitted the phase wise split up of R&M cost for the period 2022-23 which is shown below.

Table 20

Phase wise split up of R&M for the period 2022-23

Campus	Contractor	From	To	Contract Value (Rs)
Phase-I Kazhakuttam	Fathima Engineering Co. Pvt Ltd	01-07-2020	30-06-2023	7,93,16,774
Phase III Kazhakuttam	AEGIS Power Engineers	01-01-2019	30-06-2022	1,29,63,240
Phase III Kazhakuttam	AEGIS Power Engineers	01-07-2022	31-07-2022	43,18,181
Phase III Kazhakuttam	Geresh Electricals (P) Limited	01-08-2022	31-07-2025	1,24,18,800
Phase III & IV 110 kV Substations	Sterling and Wilson Pvt Ltd	01-01-2022	31-12-2024	1,80,53,836
Phase V Campus - Kollam	AEGIS Power Engineers	01-06-2020	30-09-2022	78,99,275
Phase V Campus - Kollam	AEGIS Power Engineers-Extn	01-10-2022	31-10-2022	2,84,700
Phase V Campus - Kollam	ABM Electricals Pvt. Ltd.	01-11-2022	31-10-2025	1,05,05,712

30. The licensee further stated in their petition that, the methodology adopted by the Commission for specifying the normative O&M costs for the Distribution licensees other than KSEB is arrived at based on the actual/approved expenses in the previous control period and by applying escalation factor. The licensee further stated that application of average escalation factors for arriving at the cost in the Regulation does not reflect the reality and broadens the gap between the real and approved costs. The licensee requested the Commission to consider the fact that the R&M contracts were awarded after a fair and transparent process of on-line tendering where there were a fair number of competitive bids for the contract and that there is little scope for reducing these costs. There is no way to compensate for the substantial gap between the actual R&M cost genuinely incurred and that allowed in the ARR. The licensee also stated that the Commission may please note that this difference in cost does not in any way reflect any defect or inefficiency on the part of the petitioner. The cost incurred is the lowest discovered through transparent bidding. The petitioner is legally bound to claim the contracted

amount that was discovered through transparent bidding process. The licensee also requested the Commission to allow them to meet the actual expenses in this respect.

31. The commission noted that as per the provision 80 (3) of the regulations, 'Any one time operation and maintenance expenditure of a special nature not in the form of routine repair and maintenance, if any, if required and is undertaken by the distribution licensee, expenses for such maintenance may be allowed by the commission after prudence check, considering the circumstances, details and justification furnished by the Distribution business/licensee for incurring such an expenditure, to the satisfaction of the commission'. This provision allows onetime expenses incurred, not in the nature of regular R&M expenses. ***Considering this, the Commission hereby approves the one-time repair and maintenance charges of Rs. 35.60 lakh together with the cost of material and spares amounting to Rs.11.16 lakh purchase for the same in addition to the norms as claimed by the licensee.*** The Commission also notes that the new campus at Pallippuram Technocity was commissioned in 2019-20 whereas the norms for O&M expenses were fixed based on trued up figures of 2018-19. As such, the O&M expenses for the Pallippuram Technocity are not reflected in the norm. The licensee submitted that Rs.30.38 lakh has been spent towards R&M expenses of Pallippuram Technocity. The Commission considers this and is of the opinion to allow these expenses over and above the norm. ***Therefore the Commission approves Rs.30.38 lakh as R&M expenses for Technocity in addition to the norms.***

32. The Commission examined the arguments of the licensee regarding the normative amount approved in the regulations, 2021. The commission had considered the revised R&M based on actual escalation rate for the year 2022-23. ***Accordingly, the Commission hereby approves Rs. 402.43 lakh (Rs. 35.60 lakh + Rs. 11.16 lakh + Rs.30.38 lakh + Rs. 325.29 lakh) as R&M expense for the year 2022-23.***

Table 21

R&M expenses approved for truing up 2022-23 (Rs lakh)

Particulars	As per Regulations	As per Truing Up Petition	Trued Up
R&M expenses	317.37	447.65	402.43

Administration and General Expenses

33. The licensee has claimed Rs.92.05 lakh as the A&G expenses for the year 2022-23 as against the approved norm of Rs. 36.09 lakh. The major expenses booked under A&G expense are towards conveyance charges, electricity charges, security arrangements, Electricity Duty under Section 3 of the Kerala

Electricity Duty Act, 1963 and license fee and other related fees. The claim includes Rs. 36.98 lakh paid towards Electricity Duty under section 3(1) of the Duty Act 1963. The details of the A&G expenses of truing up claim of current year and previous year are given below.

Table 22

A&G Expenses over the years (Rs. lakh)

Particulars	2021-22 Truing Up claim	2022-23 Truing up claim
Insurance	3.21	2.89
Telephone & Postage, etc.	0.37	0.35
Audit Fees	0.36	0.66
Conveyance Charges (Hired Vehicle for Distribution Business)	6.72	5.52
Electricity charges	8.47	12.10
Professional & Consultancy Charges	0.05	1.00
Bank Charges	2.74	1.89
Petition and other Fees to KSERC / KSEB	2.92	1.29
License Fee and related fee to KSERC	1.37	1.38
Security arrangements (1 person night duty per station)	11.17	11.73
Others Fees (To Electrical Inspectorate and KSEBL)	25.21	14.80
Subscription to Institution	0.00	1.18
Staff Training Expense	0.36	0.28
Allowance for Doubtful debts	0.00	-
Gross A&G Expenses	**62.95**	**55.07**
Ele. Duty u/s 3(I), KED Act	29.08	36.98
Less: Expenses Capitalised	0.00	-
Net A&G Expenses	**92.03**	**92.05**

34. As compared to the actual claim of previous year there is only a slight increase in A&G Expenses. Some of the major expenses are Electricity charges (Rs. 12.10 lakh), Fees to the Kerala State Electrical Inspectorate for periodical inspection (Rs. 14.80 lakh), the expense towards conveyance charges (Rs. 5.52 lakh) for vehicle hired exclusively for the purpose of distribution business and is used for travelling regularly to different locations within Campuses and between campuses.

35. The licensee has claimed an amount of Rs. 12.10 lakh as electricity charges. The licensee has stated that the office functions of the Electricity Distribution Business and other business activities of the Technopark is carried out from the Head Office Building called “Park Centre”. Electricity used for the administration offices are measured using separate meters. Forty percent (40%) of electricity charges for administrative offices are apportioned to the Distribution business activity. This is the same proportion taken in truing up petitions of previous years. Balance 60% is apportioned to main business activity and that amount will be revenue from sale of power for the distribution business. Regarding the expenses towards providing Security Arrangements

the licensee claimed Rs. 11.73 lakh and stated that there is only a nominal 5% increase from the amount claimed during the previous year (Rs. 11.17 lakh), the same amount was allowed in the truing up order. The total fee paid to Electrical Inspectorate is Rs. 14.80 lakh compared to Rs 25.21 lakh and Rs 17.86 lakh in previous years. This is the highest component of A&G Expenses. The fees are paid as per the demand notice issued by the inspectorate based on their inspection schedule.

36. The Commission examined the submission of the licensee and observed that, the licensee has booked Rs. 36.98 lakh towards Electricity Duty for the year 2022-23. The Commission in the previous Orders had stated that duty under Section 3 cannot be passed on to the consumers in view of the statutory provisions of the Kerala Electricity Duty Act, 1963. Hence the same is disallowed.

37. ***Considering the methodology in approving the operation and maintenance expenses based on the actual escalation for the year, the commission hereby approves the amount of Rs. 36.99 lakh as A&G expenses for the year 2022-23***

Summary of O&M Expenses approved for 2022-23

38. The summary of O&M expenses approved by the Commission for the year 2022- 23 is shown below: -

Table 23
O&M expenses for the year 2022-23 (Rs. In lakhs)

O&M Expenses	2022-23		
	As per Regulation	As per Truing Up Petition	Trued Up
Employee expenses	27.14	54.54	27.82
R&M expenses	317.37	447.65	402.43
A&G expenses	36.09	92.05	36.99
Total	**380.60**	**594.24**	**467.24**

Assets Addition

39. The licensee in the petition has stated that, there was no addition to the fixed assets during the year. There is a capital expenditure of Rs 158.27 lakh towards various on-going capital works. The licensee has clarified that these expenses are not reflected in computation until capitalized.

Table 24
Capital expenditure during 2022-23

Project / Work	Rs.lakh
HT Distribution Network - Phase-4 (Paid to KSITIL)	3.36
Solar Plant over EV Charging station-Phase-1	9.17

Solar Plant 25 kW – Ad Charge- Phase-1	0.03
Renewal of Power Distribution - Phase-1	145.49
Solar Plant 25 kW – Ad Charge - Phase-4	0.22
Total	**158.27**

40. After completion of the projects and commencement of commercial operation, the licensee has to file the details of capital investment and the Commission will approve the same after prudence check while truing up the accounts

Depreciation

41. The depreciation claimed by the licensee in the petition for truing up for the year 2022-23 is Rs.162.45 lakh. The licensee stated that no depreciation expenses for assets from Government grants and consumer contributions are claimed. The licensee has also stated that there was no asset capitalisation during the year and there is no change in the GFA in FY 2022-23. The licensee has further stated that assets worth Rs 1.73 lakh has been withdrawn from the assets which was the amount spent on electrification for rehabilitating the residents in the Technocity area, inadvertently got included in the Technocity asset computation in previous year.

Table-25
Depreciation claimed for 2022-23

Asset categories	GFA as on 1/4/2022	Asset Addition during the Year	GFA as on 1/4/2022	Accumulated Depreciation as on 1/4/2022	Rate	Depreciation for 2022-23	Accumulated depreciation as on 31/3/2022	NFA at the end of the year
	Rs. lakh	Rs. lakh	Rs. lakh	Rs. lakh	%	Rs. lakh	Rs. lakh	Rs. lakh
Land & Rights		-						
Civil works	188.95		188.95	80.59	3.34%	6.31	86.90	102.04
Transmissionlines	661.95		661.95	456.66	5.28%	34.95	491.61	170.34
Substations	2561.64		2561.64	1501.79	5.28%	135.16	1636.81	923.10
11 kV works	154.78		154.78	76.57	5.28%	8.17	84.74	70.04
LT lines/others*	73.65		73.65	26.75	5.28%	3.89	30.64	43.01
Metering equipment	6.94		6.94	3.31	5.28%	0.37	3.68	3.26
Electrical installations	96.86		96.86	29.04	5.28%	5.11	34.15	62.71
Softwares	6.26		6.26	1.86	15.00%	0.94	2.80	3.46
Total	**3751.03**	**-**	**3751.03**	**2176.57**		**194.91**	**2371.34**	**1377.96**
(Less)Assets created from Govt. grant/contributions	**612.06**		**612.06**		**5.28%**	**(32.32)**		
Depreciation for the year						**162.45**		

42. The Commission vide Order dated 16.02.2022 while truing up the accounts for the year 2019-20 (OP 48/2021) had approved a depreciation of Rs.142.60 lakh. While truing up the accounts for the year 2020-21 and 2021-22, the Commission had provisionally approved Rs.142.60 lakh, the same as

approved in 2019-20, directing the licensee to submit the complete information on the Fixed Assets indicating therein, the Assets category, procurement date and cost, salvage value, useful life of assets, capitalisation date, completed years as on 01-04-2020, remaining life of the assets, depreciation claimed till date etc. with two category viz. (a) Assets below 12 years of age and (b) Assets with more than 12 years of age. The licensee in compliance to the directions issued by the Commission, vide letter No.ETPK/FIN/KSERC/2023-24/1441 dated 27.10.2023 has submitted the details.

43. As per the submission, the licensee during the year 2020-21 has capitalized assets worth Rs.1986.98 lakh out of which an amount of Rs. 1663.14 lakh was received as grant from Government for construction of 110 kV substation in Technocity and balance Rs.323.84 lakh worth of assets was created out of own fund. The split up details is shown below.

Table-26
Assets capitalized during 2020-21 (Rs.lakh)

No	Particulars	Grant	Own fund
1	Air Insulated Substation - Technocity (110 kV)	1663.14	32.21
2	**Electrical Substation – Technocity (11 kV)** a. 11Kv USS for Technocity b. Site Development for 110kV Substation, Technocity		 51.03 228.38
3	Solar Power Plant Phase III		5.75
4	Power Distribution System		3.74
5	Billing software		1.41
6	160 KVA Outdoor Substation Phase III		1.32
	Total	**1663.14**	**323.84**

Item 1: Air Insulated Substation - Techno city (110 kV)

Technopark issued work Order for the construction of a 110kV air Insulated Substation for the Technocity campus to M/s ABB with a work order value of Rs. 1611.14 lakh. The project was carried out under the consultation of M/s KITCO Ltd, which provided consultancy services for the Supply, Installation, Testing & Commissioning of the 110kV Substation and the design of the internal power distribution within the Technocity campus, with a consultancy charge of 2.40% of the executed value of the works. The total executed value of the project amounts to Rs. 1,695.35 lakh out of which Rs. 1663.14 lakh were received from grants and the remaining Rs. 32.21 lakh were funded internally.

Item-2: Electrical Substation.

a. **11 kV USS for Technocity – Rs. 51.03 lakh**: Technopark had installed two 500kVA 11kV Unitized Substations (USS) including 11kV VCB Panel, 500kVA transformer and LT panels at Technopark Phase-IV campus (Technocity) to

provide power supply for the construction and site development projects. This installation was necessary because there was no alternate facility available for providing power on the site. This was executed by M/s Technopark Engineers India Pvt Ltd. After the site development and main substation installation work, one of the USS units was relocated to Technopark Phase-I campus to replace one of the damaged transformers in Phase-I campus, while the other remains at Technocity.

b. Site Development for 110 kV Substation, Technocity - Rs 228.38 lakh

This cost is for the site development work for the installation of 110kV substation in the Technocity campus. This work was awarded to M/s Toppil Construction with a work order value of Rs. 224.46 lakh, and the total executed work value comes to Rs. 228.38 lakh.

Item-3: Solar Power Plant Phase III - Rs. 5.75 lakh.

The amount claimed is the bill settlement for the 200 kW solar power plant installed at the roof top of Ganga / Yamuna buildings, Phase III campus. As per the payment terms in the final 5% of work order value release only after successful completion of one year monitoring of PV power plant. The power plant was commissioned on 26th March 2018 and installed by the lowest bidder (M/s KELTRON) in the competitive tender.

Item-4 Power Distribution System - Technocity - Rs 3.74 lakh

An amount of Rs.2.01 lakh was paid to KSEBL for the temporary power connection of Technocity and an amount of Rs.1.73 lakh was paid to KSEBL for the providing LT Connection to Technocity rehabilitation area

Item-5: Billing software - Rs 1.41 lakh

Technopark has already introduced a billing software called Smart Joules, developed by Ekaart Digital Systems Inc. Smart Joules is capable of handling the entire process, from meter reading entry to the generation of bills for various tariffs. The existing billing software was modified to include consumers in Kollam and Technocity Campuses, as well as to incorporate additional features for generating monthly summary tables and graphs for billing, entering power purchase bills from KSEBL, generating solar bills, and creating various reports required for submissions

Item-6: 160 KVA Outdoor Substation Phase III (Relocation works) - Rs 1.32 lakh

A 160kVA outdoor substation in the Technopark Phase-III campus was relocated from the land that was allotted to M/s Taurus to a new location near overhead water tanks in Technopark Phase-III. The earthing for the 160kVA outdoor substation and 11kV RMU relocated from the land allotted to

M/s Taurus was done temporarily. Accordingly the work is done for earthing the 160kVA substation permanently. Work Order was issued to M/s High Power Electro Mechanical Contractors for Installation of Earth Pits and electrical inspectorate electrical scheme approval and sanction for energization of relocated 160kVA outdoor substation in Technopark Phase III Campus for Rs.1.32 lakh.

44. During 2021-22, assets worth of Rs.1838.66 lakh has been capitalized which was entirely amount received as Government Grant. The amount was paid to KSEB Ltd for laying of 110 kV underground cable laying from KSEB Pothencode substation to Technocity 110kV substation.

Table-27

Assets capitalized during 2021-22 (Rs. In lakh)

Particulars	Grant
Paid to KSEB Ltd for laying of 110 kV underground cable laying from KSEB Pothencode substation to Technocity 110kV substation.	1838.66
Total	**1838.66**

45. Considering the submissions made by the licensee, the Commission approves the capital investments made by the licensee. The Commission allows to meet the own fund requirements (excluding grant received from government) of Rs. 322.11 lakh (Rs.323.84 lakh - Rs.1.73 lakh) from the approved regulatory surplus of Rs.2267.07 lakh as on 31-03-2020 vide truing up order for the year 2019-20 dated 16-02-2022 (OP No.48/2021). The Commission has been allowing all reasonable and prudent expenses including RoE/ Return on net fixed assets to Technopark over the years for carrying out its electricity distribution activities. The accrued regulatory surplus approved and allowed to be retained with the Licensee is the amount arrived at after allowing all the prudent expenses including statutory returns. Accordingly, the licensee shall not be eligible for depreciation, RoE/RoNFA and other financing costs on the assets created by utilising such surplus as it would result in double payment by the consumers.

46. Further the commission has been allowing depreciation provisionally for the years 2020-21 and 2021-22 of Rs. 142.60 lakh, (the amount approved during the year 2019-20). The licensee has now submitted the details of depreciation for the assets for the years from 2019-20 to 2022-23. Considering the submissions made by the licensee and the analysis of the commission the final calculation of depreciation is as follows:

Table 28
Depreciation for the year 2019-20

Asset categories	GFA as on 1/4/2019	Asset Addition during the Year	GFA as on 1/4/2020	Accumulated depreciation as on 1/4/2019	Depreciation rate	Depreciation for the year 2019-20	Accumulated depreciation as on 31/3/2020	NFA at the end of the year
	Rs. lakh	Rs. lakh	Rs. lakh	Rs. lakh	%	Rs. lakh	Rs. lakh	Rs. lakh
Land & Rights								
Civil works	188.95	-	188.95	67.98	3.34%	4.64	72.62	116.33
Transmission lines	661.95	-	661.95	360.04	5.28%	24.68	384.72	277.23
Substations	2244.96	-	2244.96	1087.42	5.28%	89.11	1176.53	1068.43
11 kV works	97.41	57.37	154.78	50.13	5.28%	3.27	53.40	101.38
LT lines/ others*	44.59	29.07	73.65	15.13	5.28%	1.73	16.86	56.79
Metering equipment	6.94	-	6.94	2.29	5.28%	0.31	2.59	4.35
Electrical installations	91.11	-	91.11	11.40	5.28%	2.59	13.99	77.12
Softwares	0.47	4.39	4.86	0.03	6.33%	0.17	0.20	4.66
Total	**3336.37**	**90.84**	**3427.20**	**1594.42**		**126.49**	**1720.91**	**1706.29**
(Less) Assets created from Govt. grants /contributions (Up to 2019-20)	612.06		612.06		5.28%	-32.32		
Approved Depreciation						**94.17**		

Table 29
Depreciation for the year 2020-21

Asset categories	GFA as on 1/4/2020	Asset Addition during the Year	GFA as on 1/4/2021	Accumulated depreciation as on 1/4/2020	Depreciation rate	Depreciation for the year 2020-21	Accumulated depreciation as on 31/3/2021	NFA at the end of the year
	Rs. lakh	Rs. Lakh	Rs. lakh	Rs. lakh	%	Rs.lakh	Rs. lakh	Rs. lakh
Land & Rights								
Civil works	188.95		188.95	72.62	3.34%	4.64	77.26	111.69
Transmission lines	661.95		661.95	384.72	5.28%	24.68	409.40	252.55
Substations	2244.96	1978.09	4223.05	1176.53	5.28%	89.11	1265.64	2957.41
11 kV works	154.78		154.78	53.40	5.28%	4.78	58.18	96.60
LT lines/ others*	73.65		73.65	16.86	5.28%	2.50	19.36	54.29
Metering equipment	6.94		6.94	2.59	5.28%	0.31	2.90	4.04
Electrical installations	91.11	5.75	96.86	13.99	5.28%	2.59	16.58	80.29
Softwares	4.86	1.41	6.27	0.20	6.33%	0.31	0.51	5.76
Total	**3427.20**	**1985.25**	**5412.45**	**1720.91**		**128.91**	**1849.82**	**3562.63**
(Less) Assets created from Govt. grants/ contributions (upto 2019-20)	612.06		612.06		5.28%	-32.32		
Asset addition in the year 2020-21 Accumulated surplus : Rs. 322.11 lakh Grant: Rs. 1663.14 lakh	1,985.25							
Approved Depreciation						**96.59**		

Table 30

Depreciation for the year 2021-22

Asset categories	GFA as on 1/4/2021	Asset Addition during the Year	GFA as on 1/4/2022	Accumulated depreciation as on 1/4/2021	Depreciation rate	Depreciation for the year 2021-22	Accumulated depreciation as on 31/3/2022	NFA at the end of the year
	Rs. lakh	Rs. Lakh	Rs. lakh	Rs. Lakh	%	Rs. lakh	Rs. lakh	Rs. lakh
Land & Rights								
Civil works	188.95		188.95	77.26	3.34%	4.64	81.90	107.05
Transmission lines	661.95	1838.66	2500.61	409.40	5.28%	9.73	419.13	2081.48
Substations	4223.05		4223.05	1265.64	5.28%	83.00	1348.64	2874.41
11 kV works	154.78		154.78	58.18	5.28%	4.78	62.97	91.82
LT lines/ others*	73.65		73.65	19.36	5.28%	2.50	21.86	51.80
Metering equipment	6.94		6.94	2.90	5.28%	0.31	3.20	3.74
Electrical installations	96.86		96.86	16.58	5.28%	2.55	19.13	77.73
Softwares	6.27		6.27	0.51	6.33%	0.31	0.81	5.46
Total	**5412.45**	**1838.66**	**7251.11**	**1849.82**		**107.82**	**1957.64**	**5293.47**
(Less) Assets created from Govt.grants/ contributions (upto 2019-20)	612.06		612.06		5.28%	-32.32		
Asset addition in the year 2020-21 *Accumulated surplus : Rs. 322.11 lakh Grant : Rs. 1663.14 lakh*	1985.25							
Asset addition in the year 2021-22 *Grant : Rs. 1838.66 lakh*	1838.66							
Approved Depreciation						**75.50**		

Table 31

Depreciation for the year 2022-23

Asset categories	GFA as on 1/4/2022	GFA as on 1/4/2023	Accumulated depreciation as on 1/4/2022	Depreciation rate	Depreciation for the year 2022-23	Accumulated depreciation as on 31/3/2023	NFA at the end of the year
	Rs. lakh	Rs. lakh	Rs. Lakh	%	Rs. lakh	Rs. lakh	Rs. lakh
Land & Rights							
Civil works	188.95	188.95	81.90	3.34%	4.64	86.54	102.41
Transmission lines	2500.61	2500.61	419.13	5.28%	9.73	428.86	2071.74
Substations	4223.05	4223.05	1348.64	5.28%	83.00	1431.64	2791.41
11 kV works	154.78	154.78	62.97	5.28%	4.78	67.75	87.03
LT lines/ others*	73.65	73.65	21.86	5.28%	2.50	24.36	49.30
Metering equipment	6.94	6.94	3.20	5.28%	0.24	3.44	3.50
Electrical installations	96.86	96.86	19.13	5.28%	2.55	21.68	75.18
Softwares	6.27	6.27	0.81	6.33%	0.31	1.12	5.15
Total	**7251.11**	**7251.11**	**1957.64**		**107.75**	**2,065.39**	**5185.72**

(Less) Assets created from Govt.grants/ contributions (upto 2019-20)	612.06	612.06		5.28%	-32.32		
Asset addition in the year 2020-21 *Accumulated surplus : Rs. 322.11 lakh Grant : Rs. 1663.14 lakh*	1,985.25						
Asset addition in the year 2021-22 *Grant : Rs. 1838.66 lakh*	1,838.66						
Approved Depreciation					**75.43**		

47. ***Accordingly, the Commission hereby approves depreciation of Rs.94.17 lakh for the year 2019-20, Rs. 96.59 for the year 2020-21, Rs. 75.50 lakh for the year 2021-22 and Rs. 75.43 lakh for the year 2022-23 as per the revised calculation as shown above.***

48. The Commission has allowed provisional depreciation over the past few years. Upon revising the calculations, the actual depreciation approved is lower. A comparison of the figures is provided below:

Table 32
Approved depreciation and claw back (Rs. In lakh)

Year	Depreciation originally claimed by the licensee	Depreciation approved by the commission	Revise calculated depreciation	Excess allowed in previous orders
2019-20	189.38	142.60	94.17	48.43
2020-21	154.06	142.60*	96.59	46.52
2021-22	162.68	142.60*	75.50	67.61
Total				**162.56**
* Provisionally approved				

49. ***Accordingly, the Commission hereby claws back Rs. 162.56 lakh as excess amount of depreciation provisionally approved in previous years.***

Interest and Finance Charges

50. The interest and finance charges claimed by the licensee for the year 2022-23 include interest on long term loans and interest on consumer security deposit amounting to a total of Rs.81.66 lakh as shown below.

Table-33
Interest and finance charges (Rs. lakh)

Particulars	2020-21		2021-22		2022-23
	Truing Up Petition	Trued Up	Truing Up Petition	Trued Up	Truing Up Petition
Interest and finance charges on long term loans	30.08	17.32	24.96	7.87	8.96
Interest on consumer security deposits etc	98.85	98.85	98.80	98.80	72.70
Total	**128.93**	**116.17**	**123.76**	**106.67**	**81.66**

Interest on Loan

51. The licensee has claimed interest on long term loan based on the weighted average interest rate for the year 2022-23 as shown in the Table below.

Table 34
Weighted Average Interest for the year 2022-23

No	Bank	Period of year	Rate %	Days	Weighted Rate
1	South India Bank	01-04-2022 to 01-06-2022	8.25	62	511.5
2	South India Bank	02-06-2022 to 01-12-2022	8.20	183	1500.6
3	South India Bank	02-12-2022 to 31-03-2023	8.95	120	1074.0
4	Federal Bank	01-04-2022 to 15-09-2022	8.3	168	1394.4
5	Federal Bank	16-09-2022 to 31-03-2023	8.7	197	1713.9
6	**Total**			**730**	**6194.4**
	Weighted Average Interest Rate (%)		**6194.4/730 =**		**8.49**

52. Considering the above interest rate the licensee has claimed an amount of Rs.8.96 lakh as interest on normative loan for the year 2022-23, which is tabulated below;

Table 35
Interest on normative loan claimed for the year 2022-23

Particulars	Rs lakh
Gross Loan -Opening	370.26
Cumulative repayments of Loans up to previous year	370.26
Net loan-Opening	203.91
Add: Drawal (s) during the Year	-
Less: Repayment (s) of Loans during the year	196.64
Net loan - Closing	7.27
Average Net Loan	105.59
Interest on Loan	**8.96**
Weighted average Rate of Interest on Loans	8.49%

53. Considering the decisions mentioned in para 45 to para 48 of this order the revised calculation of normative loan from the year 2019-20 to 2022-23 is as shown below:

Table 36
Revised Interest charges on normative loan (Rs. lakh)

No.	Particulars	2019-20	2020-21	2021-22	2022-23
1	Opening Normative loan	313.22	309.88	213.29	137.79
2	Add. Asset addition Particulars for the year	90.83	-	-	-
3	Less: Depreciation for the year	94.17	96.59	75.50	75.43
4	Closing Normative Loan	309.88	213.29	137.79	62.36
5	Average Loan	311.55	261.59	175.54	100.08
6	Rate of Interest (%)	10.08%	9.11%	8.28%	8.49%
7	Interest charges allowable for Asset additions (5*6)	**31.40**	**23.83**	**14.53**	**8.50**

54. ***As per the above table the Commission hereby approves Rs. 8.50 lakh as interest on normative loan for the year 2022-23 and allows an extra amount of Rs. 15.61 lakh as adjustments against interest on normative loan allowed in the past years. Comparison of approved and revised calculation of interest on normative loan is given below:***

Table 37
Approved and revised Interest on normative loan (Rs. lakh)

Year	Interest on normative loan approved	Revise calculation	Adjustments to interest on normative loan
2019-20	28.96	31.40	2.44
2020-21	17.32	23.83	6.51
2021-22	7.87	14.53	6.66
Total	**69.76**	**69.76**	**15.61**

Interest on security deposits

55. The interest on security deposits paid to the consumers for the year 2022-23 as claimed by the licensee amounts to Rs.72.70 lakh as against the ARR&ERC approved amount of Rs. 63.91 lakh. ***The Commission hereby approves the actual amount of Rs. 72.70 lakh paid by the licensee as interest on security deposits for the year 2022-23.***

Interest on working capital

56. The petitioner states that the working capital requirement has increased to Rs. 1007.87 lakh due to higher sales, compared to Rs. 765.44 lakh and Rs. 752.75 lakh in the previous two years. The consumer deposit in cash is Rs. 1790.88 lakh, which was sufficient to cover the working capital requirement. ***Therefore, there were no additional borrowings required for working capital. Hence no interest on working capital is claimed by the licensee.***

Return on Net Fixed Assets

57. The licensee has claimed Return on NFA of Rs.86.60 lakh for the year 2022-23 computed as per Regulation 28 of KSERC (Terms and Conditions for Determination of Tariff) Regulations, 2021. As per Regulation 28(2) of the Tariff Regulations 2021, if there is no equity invested in the business or equity invested in the regulated business of the distribution licensee is not clearly identifiable, return at the rate of 5.50% shall be allowed on the net fixed assets at the beginning of the financial year for such regulated business. Accordingly, the licensee has claimed a RoNFA at the rate of 5.50% on the net fixed assets at the beginning of the financial year. The claimed in the petition is tabulated below:

Table 38
Return on NFA claimed for 2022-23

Particulars	(Rs Lakh)
Net Fixed Assets net of Consumer Contribution & Grants (at the beginning of the year)	1574.46
Rate of return	5.5%
Total return on net fixed assets	**86.60**

58. As per the decisions outlined in paragraphs 45 to 48 of this order, the revised calculation of RoNFA for the period from 2019-20 to 2022-23 is presented below:

Table 39
Revised Return on NFA (Rs. In lakh)

Particulars	2019-20	2020-21	2021-22	2022-23
NFA (Net of consumer contributions and grants at the beginning of the year)	929.20	925.87	829.28	753.78
Less: NFA considered for normative loan	311.55	261.59	175.54	100.08
Net NFA for RoNFA	617.65	664.28	653.74	653.70
Rate of return (%)	3%	3%	3%	5.50%
Total return on NFA	**18.53**	**19.93**	**19.61**	**35.95**

59. ***As per the above table the Commission hereby approves Rs. 35.95 lakh as RoNFA for the year 2022-23 and claws back Rs. 14.23 lakh as excess amount of RoNFA allowed in the past years. Comparison of approved and revised calculation of RoNFA is given below:***

Table 40
Approved and revised RoNFA (Rs. lakh)

Year	RoNFA approved	Revise calculation	Adjustments to RoNFA
2019-20	27.88	18.53	9.35
2020-21	23.35	19.93	3.42
2021-22	21.07	19.61	1.46
Total	**72.30**	**58.07**	**14.23**

Collection Efficiency

60. The licensee has claimed that they have taken best effort to keep all the meters in accurate working order. The defective meters are replaced when the defects are noted. The collection efficiency in the year 2022-23 is 94.35 %. The licensee also states that (1) there is no large long-standing arrears from the consumers and (2) that surcharge is levied promptly for delay in payment. ***The Commission directs that licensee shall keep all records as per the relevant Tariff Regulations and also maintain the collection efficiency up to the optimum level for the smooth running of distribution business.***

Revenue from Sale of Power

61. The total revenue for the year 2022-23 is Rs. 5750.11 lakh for the sale of 643.34 lakh units. The sale and the revenue realised is higher compared to that of 2021-22. During 2021-22, the total revenue realised was Rs.4317.28 lakh for the sale of 500.77 lakh units. The major revenue is realised from LT IV B and HT I category of consumers. The details of revenue from sale of power are shown in below table.

Table 41
Revenue from sale of Power for the year 2022-23

Particulars	Number of consumers	Sale Lakh Units	% of total Unit sold	Revenue Rs lakh	Avg. realisation per KWh
LT – IV B IT/ITES below 10kW	220	10.93	3.57%	74.60	6.82
LT – IV B IT/ITES above 10kW up to 20kW	74	7.01	2.29%	60.35	8.61
LT – IV B IT/ITES above 20kW	238	249.31	81.43%	2418.14	9.70
LT - VI (A) General	2	0.39	0.13%	3.18	8.22
LT VI (B) General	9	0.59	0.19%	5.61	9.50
LT VI (c) General (1ph or 3 ph)	33	2.38	0.78%	25.85	10.85
LT VI (F) General (1 ph)	51	0.68	0.22%	6.20	9.07
LT VII (A) Commercial (1-Ph)	50	0.44	0.14%	4.39	10.06

LT VI (F) General (3 ph)	125	22.86	7.47%	239.51	10.48
LT VII (A) Commercial (3-Ph)	57	3.77	1.23%	47.71	12.64
LT VII (B) Commercial	11	0.02	0.01%	0.19	8.27
LT IV A Industrial below 10 KW	6	0.22	0.07%	1.33	6.16
LT IV A Industrial above 10 KW below 20 KW	4	0.38	0.12%	2.64	7.00
LT IV A Industrial above 20 KW	7	2.82	0.92%	27.19	9.63
LT VI (G) General	1	0.07	0.02%	0.69	10.49
LT IX Display Lighting and Hording	2	0.08	0.03%	1.14	14.45
Self Consumption	2	2.74	0.89%	30.26	11.05
Street Lighting	18	1.46	0.48%	6.86	4.71
Sub Total (LT)	910	306.15	100.00%	2955.83	9.65
HT -I (B) IT & IT Services	17	313.82	93.07%	2575.50	8.21
HT -II (A) General	2	13.98	4.15%	116.11	8.30
HT II B	3	5.40	1.60%	57.02	10.56
HT- IV Commercial	1	3.99	1.18%	38.49	9.66
Sub Total (HT)	23	337.19	100.00%	2787.13	8.27
Grand Total	933	643.34		5742.96	8.93
Electricity Duty Recovery				260.40	
Gross Revenue From Sale of Power				6003.37	
Less: Electricity Duty Payable to Govt. plus commission (Contra Entry)				253.25	
Net Revenue from Sale of Power				**5750.11**	

62. The average revenue realized per unit was Rs.8.93 against Rs.8.62 realized in the previous year. ***After examining the details furnished by the licensee, the Commission hereby approves the revenue from sale of power at Rs.5750.11 lakh for the purpose of Truing Up of Accounts for the year 2022-23.***

Non-Tariff Income

63. The non-tariff income accounted by the licensee for the year 2022-23 is Rs. 78.09 lakh which is lower than the previous year. The split-up details of non-tariff income claimed are shown below.

Table 42
Comparison of Non-Tariff Income (Rs.lakh)

Particulars	2021-22	2022-23
	Truing Up Petition	Truing Up Petition
Interest on investments, FD, Call Deposits & bank balances	48.90	41.02
Miscellaneous receipts	2.83	7.36
Commission for collection of electricity duty	1.93	2.47
Revenue from late payment surcharge	30.01	27.24
Total	**83.67**	**78.09**

64. The licensee stated that, the reduction of interest income was due to the reason of decrease in the interest rate provided by bank and KSEB Ltd when compared to previous year 2021-22. The Commission has examined the details of non-tariff income furnished by the licensee. The licensee has booked Rs. 41.02 lakh towards income from bank deposits. The cumulative surplus till 2021-22 is Rs. 2877.22 lakh. After adjusting the amount utilized for capital investment, the net cumulative surplus works out to Rs. 2555.11 lakh (Rs.2877.22 lakh – Rs.322.11 lakh).

65. The Commission in the Order has approved the asset additions and the same is ordered to be met from accumulated revenue surplus and hence notional interest is applicable only on the balance amount. Accordingly, the interest on the revenue surplus is approved at the rate of 5.10% per annum (the retail term deposit rate of SBI for a tenor of one year as on 01-04-2022) which worked out to Rs. 130.31 lakh (Rs. 2555.11 lakh x 5.10%). As the licensee has already accounted Rs. 41.02 lakh as interest from bank, the balance amount of Rs.89.29 lakh (Rs. 130.31 lakhs – Rs. 41.02 lakh) is treated as additional income under the head non-tariff income. ***Accordingly, the Commission approves the non-tariff income of the licensee for the year 2022-23 as Rs. 167.38 lakh (Rs. 78.09 lakh + Rs. 89.29 lakh) as against Rs. 78.09 lakh claimed in the petition.***

Revenue Surplus/Gap for 2022-23

66. Based on the above, the approved expenditure and revenue for the year 2022-23 after truing up is as shown below.

Table 43
Trued Up Income and Expenditure for the Year 2022-23 (Rs.lakh)

Particulars	ARR & ERC Approved	Petition for truing up	Trued Up
Cost of power purchase/ generation	4652.67	4652.88	4606.04
Employee Expenses	27.14	54.54	27.82
Administration and General	36.09	92.05	36.99
Repairs and Maintenance	317.37	447.65	402.43
Depreciation	142.60	162.45	75.43
Prior period item (Excess depreciation allowed)	-	-	(162.56)
Interest on long term loans	7.87	8.96	8.50
Interest on consumer deposits	63.91	72.70	72.70
Prior period item (adjustments of interest on normative loan)	-	-	15.61
Return on Net Fixed Assets	21.07	86.60	35.95
Prior period item (Excess RoNFA allowed)	-	-	(14.23)
Total Expenditure/ARR	**5268.72**	**5577.83**	**5104.68**
Revenue from sale of power	5751.09	5750.11	5750.11
Non-tariff income	159.35	78.09	167.38
Total Revenue / ERC	**5910.44**	**5828.20**	**5917.49**
Net Revenue Surplus/(Deficit)	**641.72**	**250.37**	**812.81**

Orders of the Commission

67. The Commission after considering the petition filed by M/s.Technopark for Truing up of accounts for the year 2022-23, and the clarifications and details provided by the licensee approves the following.

    a) **Total revenue is Rs.5917.49 lakh.**

    b) **Total expenditure is Rs.5104.68 lakh**

    c) **The revenue surplus for the year is Rs. 812.81 lakh**

    d) **The revised accumulated revenue surplus up to 2021-22 is Rs. 2555.11 lakh. Total cumulative revenue surplus till 2022-23 will be Rs. 3367.92 lakh (Rs. 2555.11 lakh + Rs.812.81 lakh).**

68. The petition is disposed of. Ordered accordingly.

Sd/-	Sd/-	Sd/-
Sri. T.K Jose	**Adv. A. J. Wilson**	**Sri. B. Pradeep**
Chairman	**Member**	**Member**

Approved for issue

C.R.Satheesh Chandran
Secretary